

# Methodology, Assumption and AI usage

## Methodology

The Excel database was cleaned using Python/pandas before being imported into Power BI. The data was checked for inconsistencies, missing values and repeated client records.

Parish names were cleaned by removing extra spaces, converting values to lowercase and using a mapping dictionary to correct misspellings and different versions of the same parish. The final values were checked in the notebook after cleaning.

Email addresses were checked using a regular expression and a **validate\_email** field was created with TRUE/FALSE results. A **Contactable** field was also created based on the available client information.

Client Key was used as the client identifier. The data showed that one Client Key could have multiple Client Account IDs, so repeated Client Keys were not treated as duplicate clients.

Unique clients were therefore calculated using **DISTINCTCOUNT(Client Key)**.

Power BI was used to create the dashboard, with filters for client type, status, subsidiary, parish, age group and gender. FUM was used to assess higher-value clients.

## Assumptions

- Client Key identifies a client, while Client Account ID identifies an account.
- Multiple accounts under one Client Key represent one client.
- Individual clients were used for the retail/individual analysis.
- Kingston, St. Andrew and St. Catherine were used for the Brik Pan Brik campaign because these were the specified target parishes.
- The supplied data represents one snapshot, so no historical or quarter-over-quarter data was created.
- A valid email means the email follows the expected format. It does not mean that the email is unique to one client.

## AI-Use Declaration

Tools used: Python/pandas, Jupyter Notebook, Excel, Power BI, Microsoft Copilot and ChatGPT.

AI was used for technical guidance, Python/pandas coding help, Power BI/DAX guidance, dashboard planning, email validation and data-cleaning guidance.

AI influenced parts of the coding and dashboard setup, but the data cleaning, campaign criteria, analysis and final checks were completed and validated by me.

The results were validated by checking the cleaned values in the notebook, reviewing the parish changes after standardization, checking Client Key and Client Account ID relationships, reviewing email validation results, and checking the Power BI calculations against the cleaned dataset.

Valid Email Check : <https://www.experian.com/blogs/data-quality/what-is-a-valid-email/>

Data Cleaning Understanding: <https://youtu.be/ozKsfsldV7M>

Refreshable Dashboard: [https://youtu.be/\\_EGKUA-2hwM](https://youtu.be/_EGKUA-2hwM)

DashBoardLink: [https://app.powerbi.com/links/dvYQ49RUAn?ctid=4f5b8d9f-2c66-4220-96c7-4999daa340b9&pbi\\_source=linkShare&bookmarkGuid=abc7f176-0fe1-4c8b-b318-b08b5f73d708](https://app.powerbi.com/links/dvYQ49RUAn?ctid=4f5b8d9f-2c66-4220-96c7-4999daa340b9&pbi_source=linkShare&bookmarkGuid=abc7f176-0fe1-4c8b-b318-b08b5f73d708)